# IV -B. TECH II-SEM

# DEPT: COMPUTER SCIENCE AND ENGINEERING

# SUB.CODE :21EC831OE, MEASURING INSTRUMENTS

# QUESTIONS BANK

## UNIT-1

1) Define measurement. Draw the block diagram of measurement system and explain the function of various blocks.
2) What are different types of errors and explain how to eliminate errors in instruments?
3) Describe the standard cell used as laboratory standards for voltage measurement. Also mention its advantages.
4) Explain how current balance is used to measure the unknown current.

## UNIT II

1) Discuss the working principle and application of a potentiometer sensor.
2) Explain the working principle of an electrical resistance thermometer
3) Design a measurement system for displacement measurement using LDR as sensor.
4) Explain about eddy current sensors in detail.

## UNIT III

1) Explain the working principle of vernier height gauge with a neat sketch and explain in detail about digital micrometers.
2) Sketch outside micrometer and explain its various parts
3) Discuss in detail device used for measuring rotational velocity.
4) Explain the working principle of a gyroscope along with an example and give its merits and demerits.

## UNIT 4

1). Discuss various types of elastic pressure sensing elements used in electrical transducers.

2). Explain different types of manometers in the measurement of pressure.

3). Explain how vibrating diaphragm and cylinder systems are designed for measurement of pressure.

4) Explain the working principle of ionization gauge.

## UNIT - V

1) Explain construction and the working principle of a Rotameter with a neat diagram.
2) Describe the principle of operation, construction details and limitations of an electromagnetic flow meter.
3) Explain Buoyancy method of density measurement with the help of a neat sketch
4) Explain the two float Viscorator used for measurement of viscocity.